

Ian CHOW
Astronomy Ph.D Student | University of Washington

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EDUCATION

PRESENT Sep. 2025	University of Washington PH.D ASTRONOMY SUPERVISOR : Prof. Mario Jurić	Seattle, WA, USA
Aug. 2025 Sep. 2023	University of Western Ontario MSc. ASTRONOMY Thesis : Orbital and Physical Properties of Decameter-Sized Earth Impactors SUPERVISOR : Prof. Peter G. Brown	London, ON, Canada
May 2023 Sep. 2018	University of Toronto HONOURS BSc. w/ HIGH DISTINCTION – ASTRONOMY & PHYSICS SPECIALIST, STATISTICS MAJOR, MATHEMATICS MINOR Astronomy Thesis : Analyzing Radial Velocity Data from the Resonant Planetary System HD 45364 SUPERVISORS : Dr. Sam Hadden, Prof. Hanno Rein Statistics Thesis : Probabilistic Dimensionality Reduction Methods for Stellar Chemodynamics SUPERVISOR : Prof. Joshua S. Speagle	Toronto, ON, Canada

ADDITIONAL RESEARCH POSITIONS

Aug. 2023 May 2023	Dunlap Institute for Astronomy & Astrophysics, University of Toronto SUMMER UNDERGRADUATE RESEARCH ASSISTANT Project : Understanding the impact of Bayesian inference on ultra-light axion limits SUPERVISORS : Dr. Keir K. Rogers, Prof. Renée Hložek	Toronto, ON, Canada
Aug. 2022 May 2022	Canadian Institute for Theoretical Astrophysics (CITA) SUMMER UNDERGRADUATE RESEARCH FELLOW Project : Modelling Migration Scenarios of Resonant Planets Using Radial Velocity Data SUPERVISORS : Dr. Sam Hadden, Prof. Hanno Rein, Prof. Norman Murray	Toronto, ON, Canada

PEER-REVIEWED JOURNAL PUBLICATIONS

FIRST AUTHOR

- Chow, I.**, Jurić, M., Jones, R.L., Kiker, K., Moeyens, J., Brown, P.G., Heinze, A.N., & Kurlander, J.A. “Predictions of Imminent Earth Impactors Discovered by LSST.” 2026, ApJ, 1001(1), 61.
Media Coverage/Press Release : Universe Today ([YouTube Interview](#), [Article](#)), [Sky and Telescope](#), [University of Washington](#)
- Chow, I.**, & Brown, P.G. “Decameter-sized Earth Impactors – II : A Bayesian Inference Approach to Meteoroid Ablation Modeling.” 2026, JGR : Planets, 131, e2025JE009392.
- Chow, I.**, & Brown, P.G. “Decameter-sized Earth impactors – I : Orbital properties.” 2025, Icarus, 429, 116444.
- Chow, I.**, & Hadden, S. “Influence of Modeling Assumptions on the Inferred Dynamical State of Resonant Systems : A Case Study of the HD 45364 System.” 2025, ApJ, 980(2), 236.

CONTRIBUTING AUTHOR

- Cheng, Q., Scolnic, D., Kurlander, J.A., **Chow, I.**, & Fernandes, M.B. “Assessing the Vera Rubin Observatory’s Ability to Discover Asteroid Impactors Before They Collide with Earth.”, 2026, AJ, 171(5), 309.

SELECTED AWARDS, SCHOLARSHIPS, FELLOWSHIPS & HONOURS

2026	Schweickart Prize Honorable Mention , honour	B612 Foundation
2025	Top Scholar Award , \$10,830 USD	University of Washington
2025-2028	NSERC Postgraduate Scholarship – Doctoral (PGS-D) , \$120,000 CAD	NSERC
2024	NASA International Space Apps Challenge Global Winner , honour	NASA
2024-2025	Ontario Graduate Scholarship (OGS) , \$15,000 CAD	University of Western Ontario

2023-2025	Western Graduate Research Scholarship , \$13,137 CAD	University of Western Ontario
2023	SURP Symposium Poster Competition Award , \$50 CAD	University of Toronto
2023	Summer Undergraduate Research Program (SURP) Award , \$9,980 CAD	University of Toronto
2022	Summer Undergraduate Research Fellowship (SURF) , \$9,500 CAD	CITA
2022	Smith Solis Research Scholarship in Astronomy and Astrophysics , \$1,250 CAD	University of Toronto
2020-2023	Dean's List Scholar , honour	University of Toronto

OBSERVING PROGRAMS

OBSERVING PROPOSALS AWARDED

2026B	Gemini Observatories Gemini South 8.1m , 9.8 hours ToO <i>Near-Infrared Spectroscopic Characterization of Sub-140 m Near-Earth Asteroids with Gemini/FLAMINGOS-2 : A Pilot Survey for the LSST Era</i>	Co-I
2026Q3	Apache Point Observatory ARC 3.5m , 1 half-night ToO <i>UW Solar System Rapid Follow-up Campaign : Imminent Impactors, Interstellar Objects, and Cataclysmic Cometary Events</i>	Co-I
2026Q3	Apache Point Observatory ARC 3.5m , 4 half-nights <i>Shape and Spin Pole Orientation of Active Small Bodies</i>	Co-I
2026Q3	Apache Point Observatory ARC 3.5m , 5 half-nights <i>Follow-up and Characterization of Rubin Discoveries by the UW Solar System Group : A Coordinated Campaign</i>	Co-I
2026Q3	Apache Point Observatory ARC 3.5m , 4 half-nights <i>Discovery and Characterization of Unusual Active Minor Planets</i>	Co-I
2026Q3	Apache Point Observatory ARC 3.5m , 4 half-nights <i>Characterising Elusive, Neglected, and Understudied tRansient Small-bodies (CENTAURS) : Photometric Properties and Activity Screening of Uncharacterised Centaurs</i>	Co-I
2026Q2	Apache Point Observatory ARC 3.5m , 1 half-night ToO <i>UW Solar System Rapid Follow-up Campaign : Imminent Impactors, Interstellar Objects, and Cataclysmic Cometary Events</i>	Co-I
2026Q2	Apache Point Observatory ARC 3.5m , 2 half-nights <i>Shape and Spin Pole Orientation of Main-belt Comets</i>	Co-I
2026Q2	Apache Point Observatory ARC 3.5m , 2 half-nights <i>Recovery and Characterization of Faint Trans-Neptunian Objects</i>	Co-I
2026Q2	Apache Point Observatory ARC 3.5m , 3 half-nights <i>Discovery and Characterization of Unusual Active Minor Planets</i>	Co-I

OBSERVING EXPERIENCE

2025-2026	Apache Point Observatory ARC 3.5m (IAU code 705), 13 half-nights, remote	Co-observer
2026	Višnjan Observatory TT1 1.0m (IAU code L01), 1 night, on-site	Co-observer

SELECTED ACADEMIC PRESENTATIONS

CONTRIBUTED CONFERENCE TALKS

Oct. 2026	58th Annual DPS Meeting , AAS Division for Planetary Sciences	Spokane, WA, USA
Jul. 2026	Asteroids Comets Meteors 2026 , Uniwersytet im. Adama Mickiewicza	Poznań, Poland
Sep. 2025	EPSC-DPS 2025 , Europlanet/AAS Division for Planetary Sciences	Helsinki, Finland
Jul. 2025	Meteoroids 2025 , Curtin University	Perth, WA, Australia
Mar. 2025	56th Lunar and Planetary Science Conference , NASA/Lunar and Planetary Institute	The Woodlands, TX, USA
May 2024	55th Annual DDA Meeting , AAS Division on Dynamical Astronomy	Toronto, ON, Canada
Aug. 2022	2022 CITA Planet Day , Canadian Institute for Theoretical Astrophysics	Toronto, ON, Canada

CONFERENCE POSTER PRESENTATIONS

Jun. 2024	2024 CASCA Annual General Meeting , University of Toronto/York University	Toronto, ON, Canada
Jun. 2023	2023 CASCA Annual General Meeting , NRC Herzberg/University of British Columbia	Penticton, BC, Canada

OTHER INVITED PRESENTATIONS

Apr. 2026	Center for Astrophysics and Cosmology Graduate Seminar , Univerza v Novi Gorici	<i>Ajdovščina, Slovenia</i>
Feb. 2026	Cosmology Group Seminar , Duke University	<i>Durham, NC, USA</i>
Jun. 2025	NASA Funding Review , NASA Meteoroid Environment Office (remote)	<i>Huntsville, AL, USA</i>
Feb. 2025	CSA Executive Committee Meeting , Canadian Space Agency (remote)	<i>Virtual</i>
May 2024	NASA Funding Review , NASA Meteoroid Environment Office (remote)	<i>Huntsville, AL, USA</i>

PUBLIC OUTREACH TALKS

Mar. 2026	"Great Balls of Fire : Meteors and Meteorites in the Past, Present and Future" , Astronomy on Tap Seattle	<i>Seattle, WA, USA</i>
Aug. 2025	"Brighter than the Sun : Fireballs in Earth's Atmosphere" , Hume Cronyn Memorial Observatory	<i>London, ON, Canada</i>
Apr. 2025	"SkyShield : Protecting Earth and Space Infrastructure From Space Hazards" , Royal Astronomical Society of Canada London	<i>London, ON, Canada</i>

TEACHING EXPERIENCE

My duties in the following course included delivering lectures, conducting in-class demonstrations, holding office hours, and proctoring, grading and reviewing exams.

2024-2025 **Astronomy 1021 : General Astronomy**, Teaching Assistant (x2) & Guest Lecturer *University of Western Ontario*

My duties in the following course included supervising lab sessions and grading lab reports.

2023-2024 **First-Year Physics Labs**, Teaching Assistant (x2) *University of Western Ontario*

My duties in the following courses included delivering in-person tutorials and help centres, running midterm viewing sessions, and proctoring, grading and reviewing exams.

2024 **Physics 1402 : Physics for Engineering Students II**, Teaching Assistant *University of Western Ontario*
2023 **Physics 1201 : Physics for the Sciences I**, Teaching Assistant *University of Western Ontario*

OTHER PROFESSIONAL EXPERIENCE

Sep. 2020 Jun. 2020	Innovere Medical SOFTWARE DEVELOPER	<i>Markham, ON, Canada</i>
	➤ Automated detection of dropouts in time-series audio data from an MRI scanner's wireless audio system using power spectrum analysis in MATLAB and Python, eliminating 20+ hours of work weekly ➤ Developed and tested TechSmart, an in-house multimedia app for patient use during MRI scans, with company's software development team	
	MATLAB Python Signal Processing	
Aug. 2019 Jun. 2019	Plantiga Technologies SOFTWARE DEVELOPER	<i>Vancouver, BC, Canada</i>
	➤ Developed methods to compute physical fitness heuristics from time-series acceleration (g-force) data, using signal processing techniques like digital filtering and convolution in Python (NumPy, SciPy, Pandas) to improve detection of foot impacts ➤ Field-tested and validated hardware such as sensor shoe insoles that track movement ➤ Acquired data from company partners such as physiotherapy clinics, universities (University of British Columbia, Simon Fraser University), and sports organizations (Houston Rockets, US Tennis Association) ➤ Wrote documentation of company products and services for clients	
	Python Signal Processing Data Analysis	
Aug. 2017 Jun. 2017	Synced Review RESEARCH INTERN	<i>Toronto, ON, Canada</i>
	➤ Conducted literature review focusing on advancements in reinforcement learning used in adversarial-search board and video game artificial intelligence programs for a company report ➤ Worked with company team to research and edit review articles on industry trends in machine learning and robotics technology	
	Machine Learning Artificial Intelligence Literature Review	

SELECTED LEADERSHIP, VOLUNTEERING & SERVICE

PRESENT Feb. 2026	Apache Point Observatory TELESCOPE ALLOCATION COMMITTEE (TAC) GRADUATE STUDENT REPRESENTATIVE ➤ Served as the University of Washington's graduate student representative for allocating time at the 3.5-meter Apache Point Observatory ARC telescope ➤ Read and evaluated observing proposals to equitably allocate 40+ nights quarterly in collaboration with faculty and postdoctoral members of the TAC	Seattle, WA, USA
Aug. 2025 Jun. 2024	Hume Cronyn Memorial Observatory OUTREACH VOLUNTEER ➤ Volunteered at astronomy Public Nights attended by 80+ visitors weekly at the University of Western Ontario's Cronyn Observatory	London, ON, Canada
May 2023 Jan. 2019	University of Toronto Academic Trivia Club VICE PRESIDENT, COMPETITOR, TOURNAMENT ORGANIZER & QUESTION WRITER/EDITOR ➤ Elected Vice President of the University of Toronto's Academic Trivia Club during the 2020-2021 and 2021-2022 academic years organizing twice-weekly practices and social events, managing club social media presence, and moderating practices and tournament games ➤ Organized and directed several collegiate and high school tournaments, including the 2021 University of Toronto Collegiate Novice and the 2022 University of Ottawa ACF Fall tournaments, played by 30+ collegiate teams in total across Canada and the U.S. ➤ Wrote and edited trivia questions across a wide range of academic disciplines (including astronomy and physics) for 2022 WORKSHOP, 2023 Canadian Novice, and 2024 MRNA III, collegiate tournaments played by 80+ teams in total across Canada, the U.S., and the U.K.	Toronto, ON, Canada

RELEVANT PROJECTS

SKYSHIELD ORRERY

2024

 spaceapp-wmpgang2024orrery.netlify.app/

An interactive, physics-based digital Solar System orrery highlighting near-Earth objects and meteoroids. Developed by myself, Dakota Cecil, Simon Van Schuylenbergh and Maximilian Vovk for the [2024 NASA International Space Apps Challenge](#) and selected by NASA as [one of 10 Global Winners](#) out of 10,000 submitted projects.

Press Release : [University of Western Ontario](#)

[HTML](#) [CSS](#) [JavaScript](#)

FASANO-FRANCESCHINI-TEST

2024

 github.com/wmpg/fasano-franceschini-test

A Python package implementing the multivariate extension of the two-sample Kolmogorov-Smirnov (K-S) statistical test described by [Fasano & Franceschini \(1987\)](#). Published as part of [Chow & Brown \(2025\)](#).

[Python](#)

SKILLS

Programming	Python (NumPy, SciPy, Astropy, Pandas, Matplotlib, Keras/TensorFlow, scikit-learn), MATLAB, R (ggplot, dplyr), HTML5 (Bootstrap), CSS, JavaScript (Node.js)
Software	L ^A T _E X, Git/GitHub, Jupyter Notebook, Anaconda, R Suite, Bash, Linux, SAOImageDS9, Microsoft Excel

OTHER AFFILIATIONS AND ORGANIZATIONAL MEMBERSHIPS

2025-PRESENT	LSST Solar System Science Collaboration
2025-PRESENT	American Astronomical Society
2025-PRESENT	Europlanet Society
2023-PRESENT	Canadian Astronomical Society